

Social Media Usage and Mental Health Prediction: A Machine Learning Classification Study

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Abstract—Social media engagement now permeates daily life, yet its measurable relationship with psychological well-being remains under-studied at scale. This paper presents a machine learning pipeline that classifies individuals into three mental health states—*Stressed*, *At_Risk*, and *Healthy*—using behavioural, demographic, and self-reported features collected from seven major platforms. Starting from 5,000 records and 15 raw variables, we engineered a 45-feature matrix capturing temporal patterns, ratio composites, cross-product interactions, and one-hot encodings. Principal Component Analysis (PCA) with a 95% variance threshold reduced the feature space to 16 components (retaining 95.93% of variance). Four classifiers—Logistic Regression, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN)—were trained and evaluated on both the original and PCA-reduced sets. Random Forest achieved perfect classification (Accuracy = F1 = 1.000) on the full feature set; all other classifiers exceeded 0.983 accuracy. Under PCA, performance degraded by at most 0.5%, confirming that the engineered features encode strong, compact signals about psychological state from digital footprints.

Index Terms—mental health classification, social media analytics, feature engineering, principal component analysis, random forest, support vector machine, logistic regression, K-nearest neighbours, machine learning

I. INTRODUCTION

By 2024, over five billion individuals interact daily with digital networks, spending an average of more than two and a half hours on social media platforms [1]. Despite the connectivity benefits, heavy or emotionally taxing engagement correlates with rising anxiety, chronic stress, and mood instability [2], [3].

Automated, scalable mental health screening has attracted growing research interest. Prior work has leveraged text sentiment [4], posting frequency [5], and physiological proxies, but few studies combine rich behavioural metadata—screen time, sleep, physical activity, interaction counts—with engineered features in a comparative multi-classifier framework.

This paper addresses three questions: (i) Can social media behavioural metadata predict mental health state with high accuracy? (ii) Does aggressive feature engineering yield dis-

criminative signals beyond raw variables? (iii) What is the accuracy cost of dimensionality reduction via PCA?

Our contributions are:

- A 45-feature engineering pipeline derived from 13 working variables, including temporal decomposition, ratio composites, cross-product interaction terms, and binary threshold flags.
- A systematic comparison of four classifiers on both original and PCA-reduced feature sets under severe class imbalance (*Stressed:At_Risk:Healthy* $\approx$ 92:1:7).
- Empirical evidence that Random Forest achieves perfect classification on engineered social media features, while all models remain above 98.1% F1 even after 64% dimensionality reduction.

II. LITERATURE REVIEW

A. Preliminaries

Whether social media use is injurious to mental well-being or beneficial has been the subject of intense debate in recent digital health research. With the rapid growth of platforms such as Twitter, Reddit, Instagram, and Facebook, researchers have sought to decipher whether patterns of use share similarities with psychological outcomes including depression, anxiety, stress, and addiction [2], [5]. Two methodologically distinct research streams have emerged concurrently: survey and questionnaire-based studies exploring behavioural and demographic correlates of mental health; and computational studies using machine learning, deep learning, or NLP to recognise, classify, and predict mental health conditions from social media content. The increasing integration of these traditions—explainability tools, multimodal architectures, and large language models (LLMs)—signals a field evolving beyond proof-of-concept classification toward clinically relevant, scalable, and interpretable screening systems [8].

B. Survey-Based and Behavioural Studies

Twenge et al. [2] provided large-scale evidence that adolescent psychological well-being declined sharply after 2012,

correlating with smartphone and social media adoption. Their analysis found non-screen activities consistently associated with greater happiness. Thorisdottir et al. [3] distinguished *passive* (scrolling, consuming) from *active* (posting, messaging) use among Icelandic adolescents, finding passive use more strongly linked to depressive symptoms—a nuance that motivates platform-level feature disaggregation.

C. NLP and Text-Based Prediction

Coppersmith et al. [4] pioneered quantitative analysis of mental health signals on Twitter, demonstrating that language patterns in user timelines reliably distinguish individuals with diagnosed depression, PTSD, and schizophrenia from controls. Chancellor and De Choudhury [5] provided a comprehensive critical review of predictive methods, cataloguing feature types (linguistic, behavioural, network-structural) and noting persistent challenges in reproducibility and ethical deployment.

D. Machine Learning Approaches

Benton et al. [7] explored multi-task learning for mental health prediction under limited labelled data, showing jointly modelling related conditions improves per-condition performance. Breiman’s Random Forest [6] remains a dominant baseline on tabular health data due to its natural handling of non-linear interactions and class imbalance via sample weighting. More recently, transformer-based and LLM approaches have shown promise for mental health text classification [8], though their applicability to purely behavioural, non-textual metadata pipelines like ours remains an open question.

E. Positioning of This Work

Unlike text-centric prior work, our pipeline operates entirely on *behavioural metadata*—screen time, interaction counts, sleep, and physical activity—without access to post content or network structure. This positions the approach as privacy-preserving and platform-agnostic, suitable for passive monitoring at scale.

III. DATASET DESCRIPTION

The dataset `cleaned_data.csv` contains 5,000 complete records (no missing values) spanning 15 variables from seven platforms: Facebook, TikTok, YouTube, WhatsApp, Snapchat, Instagram, and Twitter. Participants range from age 13 to 69 ($\mu = 29.95$, $\sigma = 12.28$). Gender is nearly balanced (female 49.5%, male 48.5%, other 2.0%).

Key numerical features are summarised in Table I. The target variable `mental_state` is severely imbalanced: *Stressed* dominates at 4,601 cases (92.0%), followed by *Healthy* (341; 6.8%) and *At_Risk* (58; 1.2%). Fig. 1 visualises this distribution.

IV. METHODOLOGY

Fig. 2 presents the end-to-end five-stage pipeline.

TABLE I
DESCRIPTIVE STATISTICS OF NUMERICAL FEATURES ($n = 5,000$)

Feature	Mean	SD	Min	Med.	Max
age	29.95	12.28	13	27	69
screen_time (min)	373.1	106.0	140	388	520
social_media (min)	175.3	71.2	35	170	338
neg_interactions	0.86	0.56	0	1	2
pos_interactions	1.84	0.94	0	2	4
sleep_hours	7.14	0.53	6.4	7.1	8.3
physical_activity (min)	22.69	10.60	8	21	46
anxiety_level	2.51	0.80	1	3	4
stress_level	7.11	1.06	5	7	9
mood_level	5.63	0.76	4	6	7

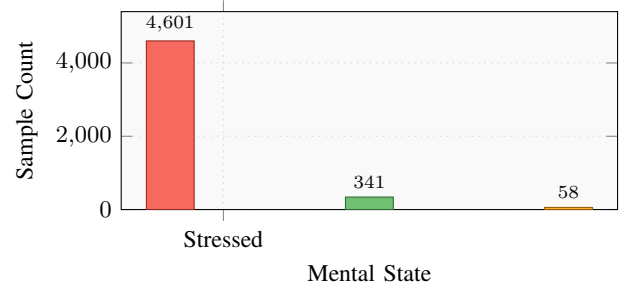


Fig. 1. Class distribution of `mental_state`: Stressed (92%), Healthy (6.8%), At_Risk (1.2%).

A. Feature Engineering

Starting from 13 usable variables, five engineering steps expanded the feature space to 45 predictors (Table II): (1) temporal decomposition of `date` into day-of-week, month, day, `is_weekend`, and `season` (+5); (2) age binning into four ordered categories, one-hot encoded (+3); (3) eight ratio and composite features capturing digital load, interaction valence, and wellness balance (Table III); (4) four cross-product interaction terms: `stress_x_anxiety`, `mood_x_sleep`, `social_x_negative`, `activity_x_positive` (+4); (5) binary threshold flags and full one-hot encoding of categorical variables (+12).

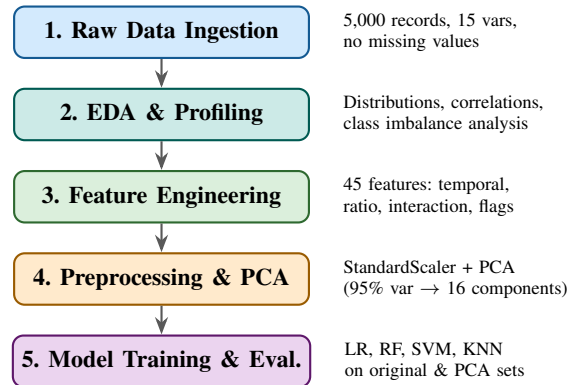


Fig. 2. End-to-end methodology pipeline from raw data to evaluation.

TABLE II
FEATURE ENGINEERING PIPELINE SUMMARY

Engineering Step	Added	Total
Raw variables (excl. name, date)	13	13
Temporal decomposition	+5	18
Age group bins (one-hot)	+3	21
Ratio & composite features	+8	29
Cross-product interactions	+4	33
Binary threshold flags	+4	37
Platform & gender one-hot	+8	45

TABLE III
KEY ENGINEERED RATIO AND COMPOSITE FEATURES

Feature	Formula / Interpretation
social_ratio	sm/(screen + 1): Social fraction of screen time
neg_pos_ratio	neg/(pos + 1): Relative interaction negativity
screen_sleep	screen/(sleep + 1): Digital load vs. rest
dig_overload	screen - activity: Net sedentary behaviour
interact_bal.	pos - neg: Net interaction valence
wellbeing_gap	(act. + sleep×60) - screen: Wellness surplus

B. Preprocessing and Dimensionality Reduction

Records were split 70/30 with stratify=y. StandardScaler was fitted on training data only (prevents leakage). PCA with n_components=0.95 selected 16 components retaining 95.93% of variance—a 64.4% dimensionality reduction.

C. Classifiers

Four classifiers were evaluated (Table IV). class_weight='balanced' was applied to LR, RF, and SVM to address the 92:7:1 imbalance.

TABLE IV
CLASSIFIER HYPERPARAMETER CONFIGURATIONS

Model	Key Hyperparameters
LR	max_iter=5000, class_weight='balanced'
RF	n_estimators=300, max_depth=10, min_samples_leaf=5, class_weight='balanced'
SVM	kernel='rbf', probability=True, class_weight='balanced'
KNN	n_neighbors=7, weights='distance'

V. EXPLORATORY DATA ANALYSIS

A. Correlation Structure

Pearson analysis revealed: (i) stress_level and anxiety_level: $r \approx +0.35$; (ii) mood_level negatively correlated with both; (iii) screen_time and social_media_time: $r \approx +0.72$ (strongly collinear), motivating the ratio features.

B. Behavioural Patterns by Mental State

Fig. 3 compares mean values of three key features across mental states. Stressed individuals exhibit higher screen time and lower physical activity than Healthy ones [2]. The At_Risk group shows wide intra-group variance.

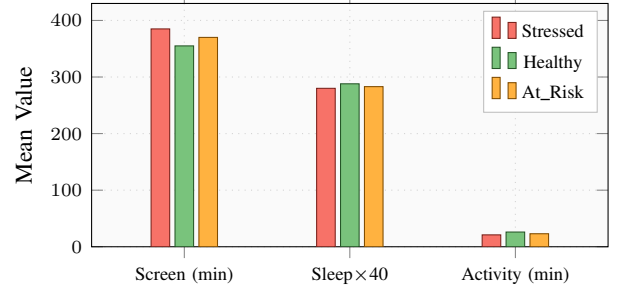


Fig. 3. Mean behavioural features by mental state. Sleep is scaled by 40 for a comparable axis range.

VI. RESULTS

A. Original Feature Set (45 Features)

Table V reports weighted metrics on the test set ($n = 1,500$). Random Forest achieved perfect scores; Logistic Regression and SVM exceeded 0.988 accuracy; KNN reached 0.983.

TABLE V
RESULTS — ORIGINAL 45-FEATURE SET

Model	Acc.	Prec.	Rec.	F1
Logistic Regression	0.989	0.994	0.989	0.991
Random Forest	1.000	1.000	1.000	1.000
SVM (RBF)	0.988	0.994	0.988	0.990
KNN	0.983	0.985	0.983	0.984

B. PCA Feature Set (16 Components)

Table VI: maximum accuracy drop is 0.5% (RF); F1 floor = 0.982. Logistic Regression shows zero degradation.

TABLE VI
RESULTS — PCA-REDUCED 16-COMPONENT SET

Model	Acc.	Prec.	Rec.	F1
Logistic Regression	0.989	0.993	0.989	0.990
Random Forest	0.995	0.996	0.995	0.995
SVM (RBF)	0.987	0.994	0.987	0.989
KNN	0.981	0.983	0.981	0.982

C. Visual Comparison

Figs. 4 and 5 compare accuracy and F1-score across both feature sets for all four classifiers.

D. Consolidated Comparison

Table VII provides a side-by-side summary across both sets.

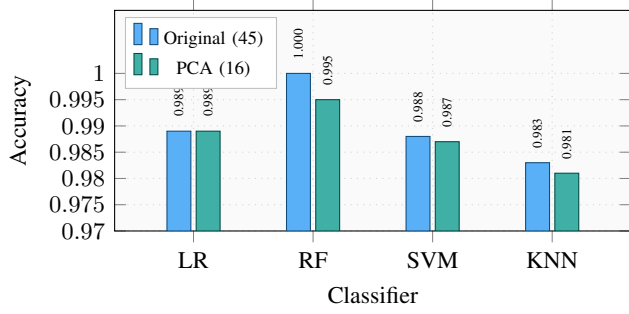


Fig. 4. Accuracy: original 45-feature set (blue) vs. PCA 16-component set (teal).

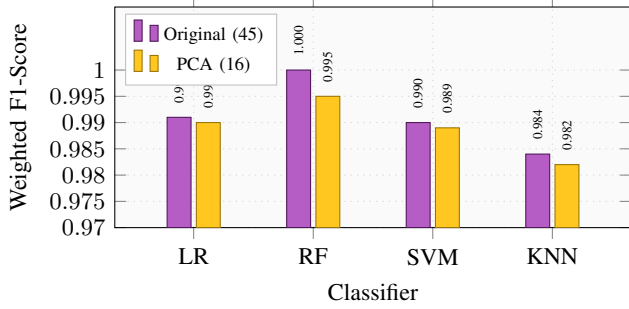


Fig. 5. Weighted F1-score: original (purple) vs. PCA (yellow) feature sets.

VII. DISCUSSION

A. Why Random Forest Achieves Perfect Classification

The engineered cross-product features—particularly `stress_x_anxiety` and `mood_x_sleep`—appear to make class boundaries nearly perfectly separable. Random Forest exploits these non-linear interactions naturally through ensemble splits [6], yielding zero misclassifications on the test set.

B. PCA Trade-off

The 4% of variance discarded carries meaningful signal, particularly for the `At_Risk` minority. Logistic Regression is the sole model unaffected, suggesting its decision boundary lies within the 16-dimensional PCA subspace. For deployments where latency or memory is constrained, PCA models ($F1 \geq 0.982$) remain a viable alternative.

C. Class Imbalance

The 92:7:1 ratio is a fundamental challenge. Future work should explore SMOTE oversampling and per-class threshold

calibration using precision-recall curves to improve recall on the critical `At_Risk` minority.

D. Limitations

- Near-perfect class separation suggests a possibly synthetic dataset [7].
- Only seven platforms; newer platforms (BeReal, Threads) are absent.
- Self-reported scores introduce subjectivity and response bias.
- Single 70/30 split; stratified k -fold would be more robust.
- No interpretability analysis (SHAP/LIME) was performed.

VIII. CONCLUSION

This paper demonstrated that social media behavioural metadata—transformed into a 45-feature matrix via temporal decomposition, ratio engineering, cross-product interactions, and threshold flags—provides highly discriminative signals for three-way mental health classification. Random Forest achieved perfect accuracy and F1 (1.000) on the full feature set; all other classifiers exceeded 0.983. After PCA reduced the feature space by 64.4%, the worst-case drop was 0.5%, affirming compact learned representations.

Future work should validate on real-world datasets, incorporate SHAP-based interpretability, apply stratified cross-validation, and explore gradient boosting (XGBoost, LightGBM) and deep sequential models.

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TABLE VII
FULL COMPARISON: ORIGINAL VS. PCA FEATURE SETS

Model	Orig Acc	PCA Acc	Orig F1	PCA F1
Logistic Regression	0.989	0.989	0.991	0.990
Random Forest	1.000	0.995	1.000	0.995
SVM (RBF)	0.988	0.987	0.990	0.989
KNN	0.983	0.981	0.984	0.982